

Student Academic Performance Analysis

Identifying Factors Affecting Exam Scores

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Program: B.Tech – Artificial Intelligence & Data Science

Project Type: Data Analytics with AI Academic Internship Project

1. Project Overview

This project analyzes student academic performance using Python and examines how academic, family, school, and social factors are associated with Exam_Score.

2. Objective

The objective is to identify factors associated with student exam scores by performing data preparation, exploratory data analysis, correlation analysis, group-wise comparisons, and visualizations.

3. Dataset Description

The dataset contains 6,607 student records and 20 columns. The target variable is Exam_Score. Numeric variables include Hours_Studied, Attendance, Sleep_Hours, Previous_Scores, Tutoring_Sessions, Physical_Activity, and Exam_Score. The dataset source is identified in the project README as Zenodo – Student Performance Factors.

4. Tools and Technologies

The project uses: Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab.

5. Data Preparation and Cleaning

The dataset was loaded with Pandas. The notebook checked shape, data types, missing values, descriptive statistics, and duplicate records. Missing values in Teacher_Quality, Parental_Education_Level, and Distance_from_Home were filled with the categorical label 'Missing'.

6. Exploratory Data Analysis

The notebook examines the dataset structure, descriptive statistics, score distribution, attendance ranges, study-hour ranges, tutoring sessions, and categorical factors such as parental involvement, access to resources, motivation, teacher quality, peer influence, internet access, family income, parental education, distance from home, and learning disabilities.

7. Correlation Analysis

The notebook calculates correlations between numeric variables and Exam_Score. The following values were obtained:

Factor	Correlation with Exam Score
Attendance	0.581072
Hours Studied	0.445455
Previous Scores	0.175079
Tutoring Sessions	0.156525
Physical Activity	0.027824
Sleep Hours	-0.017022

8. Categorical Factor Analysis

Group-wise averages were calculated for several categorical variables. Selected results are shown below.

Factor	Higher/associated group average
Parental Involvement	High: 68.09
Access to Resources	High: 68.09
Motivation Level	High: 67.70
Teacher Quality	High: 67.68
Peer Influence	Positive: 67.62
Internet Access	Yes: 67.29
Extracurricular Activities	Yes: 67.44

9. Additional Observations

Factor	Observed average-score pattern
Attendance	60–69%: 64.21; 90–100%: 70.14
Hours Studied	1–10: 63.97; 31–40: 70.98
Tutoring Sessions	0 sessions: 66.49; 5 sessions: 69.07; 6 sessions: 71.67
Family Income	High: 67.84; Medium: 67.33; Low: 66.85
Parental Education	Postgraduate: 67.97; College: 67.32
Distance from Home	Near: 67.51; Far: 66.46
Learning Disabilities	No: 67.35; Yes: 66.27

10. Key Findings

- Attendance has the strongest relationship with Exam_Score among the numeric factors examined.
- Hours Studied has a moderate positive relationship with Exam_Score.
- Previous Scores and Tutoring Sessions show weaker positive relationships with Exam_Score.
- Higher parental involvement, better access to resources, higher motivation, higher teacher quality, and positive peer influence are associated with higher average scores.
- Internet access and extracurricular activities show small differences in average scores.
- The overall average Exam_Score in the notebook is approximately 67.24.

11. Conclusion

The analysis indicates that attendance and study time are important factors associated with student academic performance. Other academic, family, school, and social factors also show differences in average exam scores. These findings describe associations in the analyzed dataset and do not by themselves establish causation.

12. Project Files

- Student_Performance_Cleaned.csv
- requirements.txt
- README.md
- student_performance_analysis.py
- project_summary.txt

13. References

Dataset source identified by the project: Zenodo – Student Performance Factors. The analysis and findings in this report are based on the submitted Google Colab notebook.